APPLICATION NOTE · ESP32S3-ITC97

Pervasive iTC 9.7″ Dual-COG Driver

Register sequences, OTP layout and short-waveform tables

PANEL	VARIANTS COVERED	FRAME BUFFER	SOURCE
960 × 672, black / white / red two COGs, one FPC	E2969JS084 — chapter 2 E2969JS0B1 — chapter 3	2 planes × 80 640 B 1 bpp, MSB first	epd97.cpp, epd97_lut.h epd97_config.h

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1 Panel

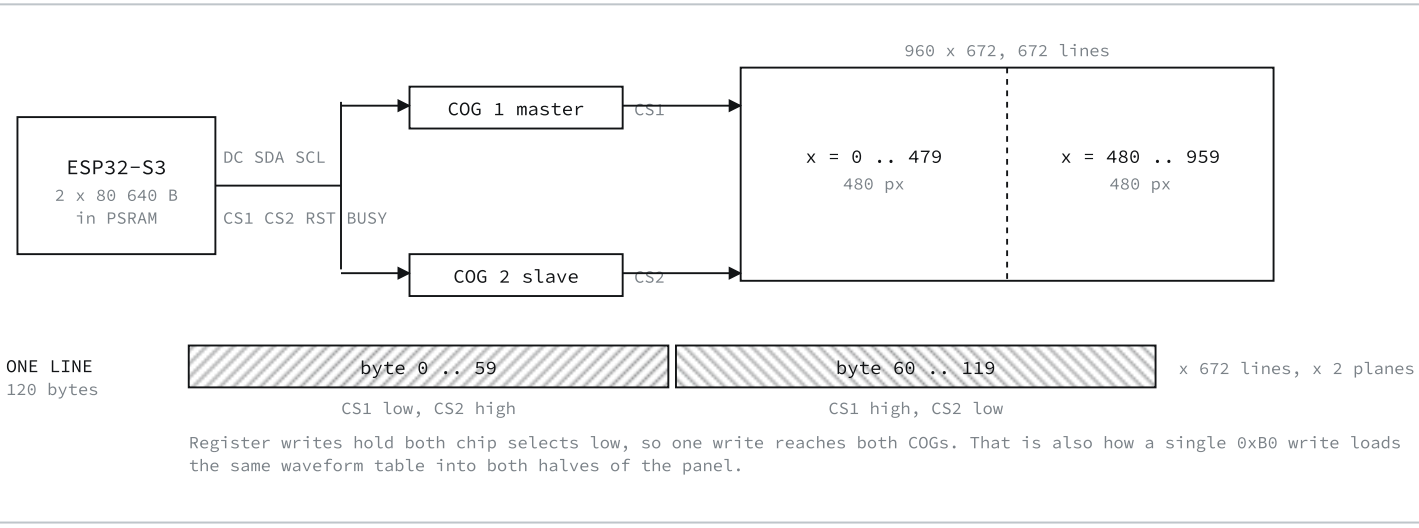
Common to both variants. Chapters 2 and 3 are self-contained and do not refer to one another.

1.1 Geometry and bus

960 × 672 pixels. Two COGs behind one FPC, each owning half of every scan line. A line is 120 bytes: 60 to the master with CS1 low, chip select swapped, 60 to the slave with CS2 low. Register writes hold both chip selects low and so reach both COGs at once.

Two buses share MOSI and SCK. Register writes and the OTP readback are bitbanged — a few dozen bytes, and the OTP needs a ~50 kHz clock with the data pin turned around for the read. The two 80 640-byte planes go out over hardware SPI at 4 MHz. The GPIO matrix is re-pointed between the two; chip select stays plain GPIO, because it is swapped every 60 bytes.

FIG. 1-1 BUS AND SCAN-LINE SPLIT



1.2 Pixel encoding

A pixel is a pair of bits, one in each plane, and the pair selects a waveform table as `(red << 1) | bw`. Red is therefore not black with red on top: a red pixel has the red bit set and the bw bit **clear**.

TABLE 1-1 PLANE BITS TO TABLE INDEX

RED	BW	TABLE	ROLE	RESULT ON THE FILM
0	0	0	paper	white
0	1	1	ink	black
1	0	2	red	red
1	1	3	unreachable	no waveform in this driver drives table 3
—	—	4	VCOM	common plane; no pixel selects it

1.3 Register index summary

Every index this driver writes. The full sequences, with per-byte meaning, are in chapters 2 and 3.

TABLE 1-2 ALL REGISTER INDICES

INDEX	FUNCTION	084	B1	PAYLOAD
0x01	DCTL	yes	yes	OTP[0x10]
0x02	last step of the init sequence	yes	yes	OTP[0x11]
0x03	oscillator	yes	no	00, OTP[0x12]
0x05	panel setting / power sequencing	yes	yes	7D, then 00
0x09	booster enable / level	yes	yes	from OTP ramp
0x10	black/white plane	yes	yes	80 640 B
0x11	red plane	yes	yes	80 640 B
0x12	RAM_RW pointer	yes	yes	OTP[0x13..15]
0x13	DUW, display update window	yes	yes	OTP[0x16..1B]
0x15	start the update	yes	yes	3C
0x44	waveform source pointer	yes	yes	00, then 06
0x45	waveform bank select	yes	yes	80, then 86 / 84
0x51	DC/DC phase voltages	yes	yes	ph0, ph1
0x60	TCON	yes	yes	OTP[0x0B]
0x61	STV	yes	yes	OTP[0x1C]
0x70	booster configuration, both COGs	no	yes	71, 36
0x90	DRFW, data refresh window	yes	yes	OTP[0x0C..0F]
0xA1	OTP unlock, step 1	yes	no	00
0xA3	OTP start address	yes	no	00, 40
0xA7	OTP / waveform load strobe	yes	yes	80, or 10 / 00
0xA8	OTP read opener	yes	no	bare index
0xB0	short waveform table	yes	yes	180 B / 400 B
0xB9	OTP read opener	no	yes	bare index
0xC2	unnamed panel setting	yes	no	3F
0xD6	unnamed panel setting	yes	yes	OTP[0x1E]
0xD8	unnamed panel setting	yes	yes	OTP[0x1D]

NOTE Indices whose function is not evidenced by the driver source are listed as unnamed rather than given a name they may not have. Their arguments are nevertheless fixed and are reproduced exactly.

2 Driver 8 — E2969JS084

Complete sequence for this variant. Identified by an OTP header of `0xA5`. Waveform bank `0x86`, 180-byte short-waveform table, refresh duration follows the table.

2.1 OTP probe

Run once per power cycle, before anything else. The result is cached and also kept in RTC memory, so a deep-sleep wake skips it. Clocked at ~50 kHz throughout.

TABLE 2-1 084 OTP READ

#	INDEX	BYTES	MEANING
1	—	—	VCC_EN on, 800 ms, then RST high 50 / low 100 / high 400 ms
2	`0xA1`	`00`	unlock, step 1
3	`0xA7`	`80`	OTP strobe
4	`0xA3`	`00, 40`	start address of the read
5	`0xA8`	—	read opener, index only, no data phase. CS1 stays low afterwards
6	—	`1 + 88 B`	SDA turned to input; one dummy byte, then the 88-byte block, MSB first
7	—	—	OTP[0x00] must read `0xA5`, else the probe failed

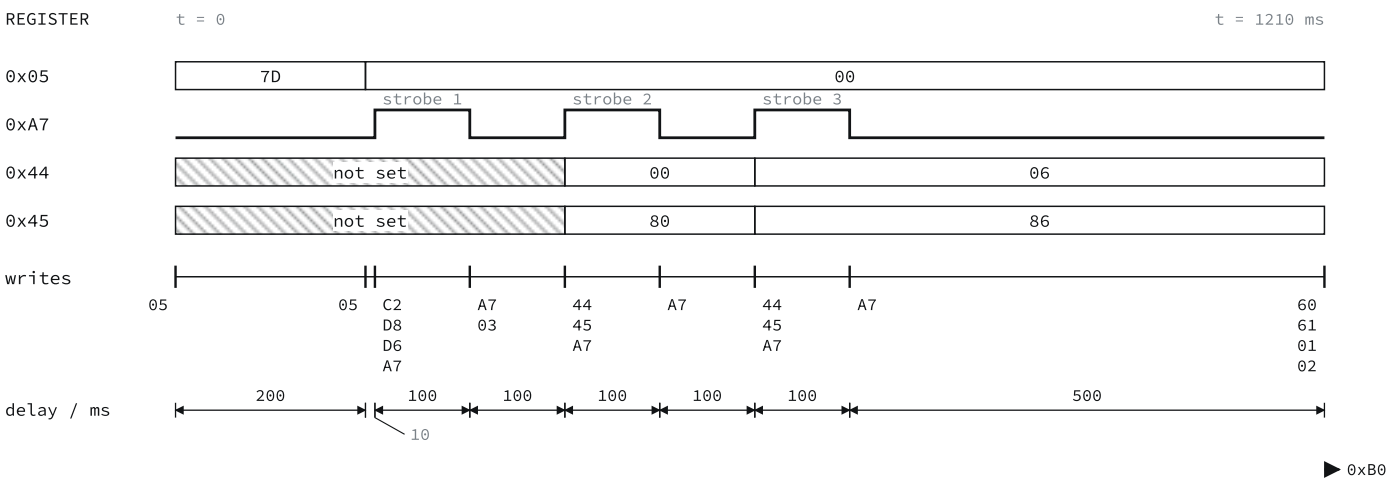
2.2 Initialisation sequence

Execution order for one image. Steps 5–10 carry the frame, steps 11–29 are the register sequence, and step 30 is the only point at which the short waveform can be written.

TABLE 2-2 084 INITIALISATION, COMPLETE

#	INDEX	BYTES	MEANING OF EACH BYTE	DELAY AFTER
POWER AND RESET				
1	—	—	VCC_EN asserted. The 1500 ms minimum supply-off time is waited out first	800 ms
2	—	—	RST high	50 ms
3	—	—	RST low	100 ms
4	—	—	RST high	400 ms
FRAME DATA				
5	0x13	OTP[16..1B]	DUW — display update window. 6 bytes, see Table 2-3	—
6	0x90	OTP[0C..0F]	DRFW — data refresh window. 4 bytes, see Table 2-3	—
7	0x12	OTP[13..15]	RAM_RW pointer. 3 bytes, forwarded verbatim	—
8	0x10	80 640 B	black/white plane. 672 lines × (60 B with CS1 low, 60 B with CS2 low)	—
9	0x12	OTP[13..15]	RAM_RW pointer again — it is re-sent before each plane, not once	—
10	0x11	80 640 B	red plane. Always sent; a frame with no red sends zeros	660 ms
REGISTER SEQUENCE — 1210 MS IN TOTAL				
11	0x05	7D	opens the sequence	200 ms
12	0x05	00	—	10 ms
13	0xC2	3F	unnamed panel setting, literal	—
14	0xD8	OTP[1D]	unnamed panel setting, from the fuse block	—
15	0xD6	OTP[1E]	unnamed panel setting, from the fuse block	—
16	0xA7	10	load strobe 1, high	100 ms
17	0xA7	00	load strobe 1, low	100 ms
18	0x03	00, OTP[12]	oscillator. Injected here, immediately after strobe 1 falls	—
19	0x44	00	waveform source pointer	—
20	0x45	80	bank, the driver's own default	—
21	0xA7	10	load strobe 2, high	100 ms
22	0xA7	00	load strobe 2, low	100 ms
23	0x44	06	waveform source pointer	—
24	0x45	86	bank. 0x86 is the 9.7″; 0x8C is the 12″	—
25	0xA7	10	load strobe 3, high	100 ms
26	0xA7	00	load strobe 3, low. This pair commits the OTP waveform	500 ms
27	0x60	OTP[0B]	TCON. Written before STV, which is what this panel size wants	—
28	0x61	OTP[1C]	STV	—
29	0x01	OTP[10]	DCTL	—
30	0x02	OTP[11]	closes the sequence	—
WAVEFORM, BOOSTER, TRIGGER				
31	0xB0	180 B	short waveform table — fast refresh only. Section 2.5	—
32	0x09 0x51	from OTP	DC/DC soft start. Four stage records at OTP[0x20..0x3F], section 4.2	100 ms
33	—	—	wait for BUSY high. Ceiling 10 s, then it proceeds anyway	—
34	0x15	3C	start the update. Sent **once** — the table sets the duration	—
35	—	—	BUSY low within 2 s, then high within 180 s	—
36	—	—	watch 1 s in case BUSY drops again for a second waveform pass	—

FIG. 2-1 084 REGISTER SEQUENCE AGAINST REAL TIME



At t = 1210 the sequence ends. Then 0xB0 takes 180 B on a fast refresh, then the DC/DC soft start, 100 ms settle, 0x15 once. Two-digit labels are register indices; each argument is in the sequence table above.

The axis is the real 1210 ms. The three load strobes appear as what they are — 100 ms high, 100 ms low, three times over — with the source pointer and the bank set in the gaps between them. The 500 ms after the last strobe is the largest single gap in the sequence.

NOTE Steps 19–26 are what load the panel's own waveform into the registers that step 31 replaces. Writing `0xB0` before step 26 gets overwritten; writing it after step 32 is too late. Step 31 is the only window.

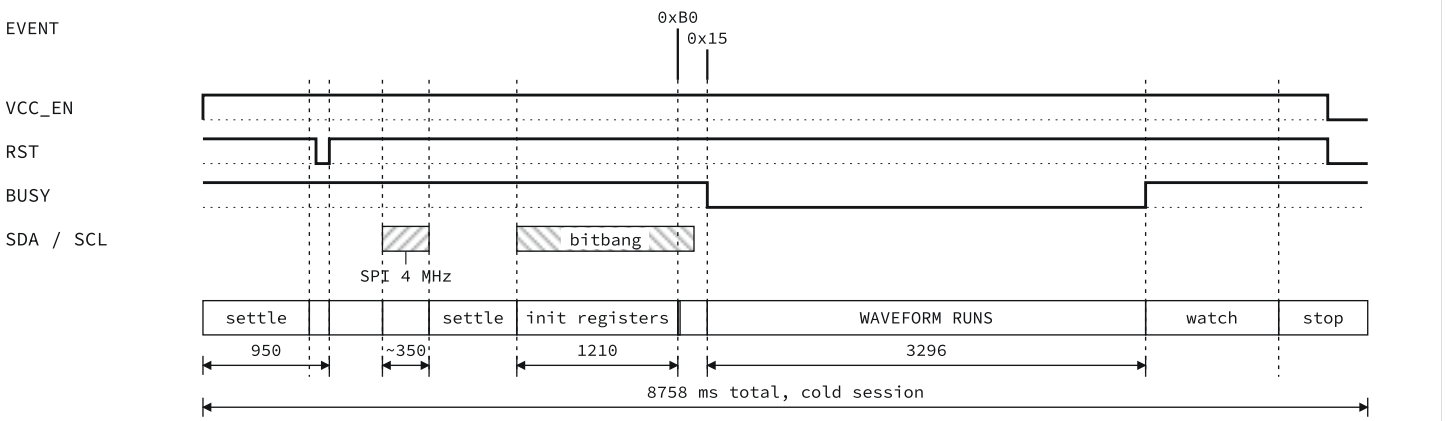
2.3 Window registers, byte by byte

TABLE 2-3 MULTI-BYTE PAYLOADS

INDEX	BYTE	SOURCE	MEANING
0x13 DUW	[0]	OTP[0x16]	x start, in bytes
	[1]	OTP[0x17]	x end, in bytes. Reads `0x3B` = 59, i.e. 60 bytes per COG. Validated
	[2]	OTP[0x18]	y window. Forwarded verbatim; the driver does not interpret these
	[3]	OTP[0x19]	
	[4]	OTP[0x1A]	
	[5]	OTP[0x1B]	
0x90 DRFW	[0]	OTP[0x0C]	forwarded verbatim
	[1]	OTP[0x0D]	x end. Reads `0x3B`. Validated
	[2]	OTP[0x0E]	forwarded verbatim
	[3]	OTP[0x0F]	
0x12 RAM_RW	[0]	OTP[0x13]	RAM read/write pointer. Forwarded verbatim, re-sent before each plane
	[1]	OTP[0x14]	
	[2]	OTP[0x15]	

2.4 Refresh timing

FIG. 2-2 084 FAST REFRESH, SUPPLY-ON TO SHUTDOWN



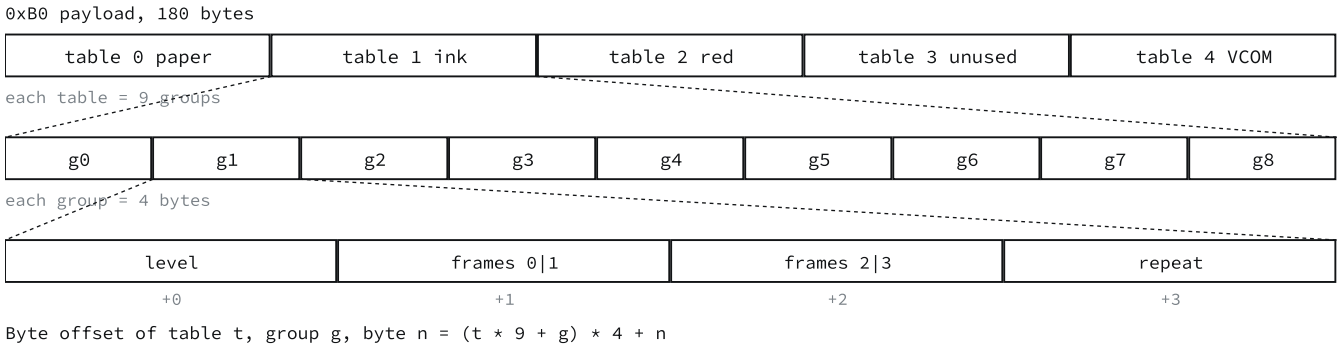
All figures in ms. A kept-open session re-enters at the plane write and ends at the trigger: no supply cycle, no reset, no register sequence, no soft start.

One trigger, and BUSY stays low for the whole 3296 ms the table asks for. A slow refresh is the same diagram with step 31 omitted and the low period at about 51 s.

2.5 Waveform table at 0xB0

180 bytes: five tables of nine groups of four bytes. Table index comes from the pixel bit pair (Table 1-1); table 4 is VCOM.

FIG. 2-3 180-BYTE PAYLOAD GEOMETRY



2.5.1 Level byte

Two bits per phase, phase 0 in the top bits. The same encoding is used by both variants.

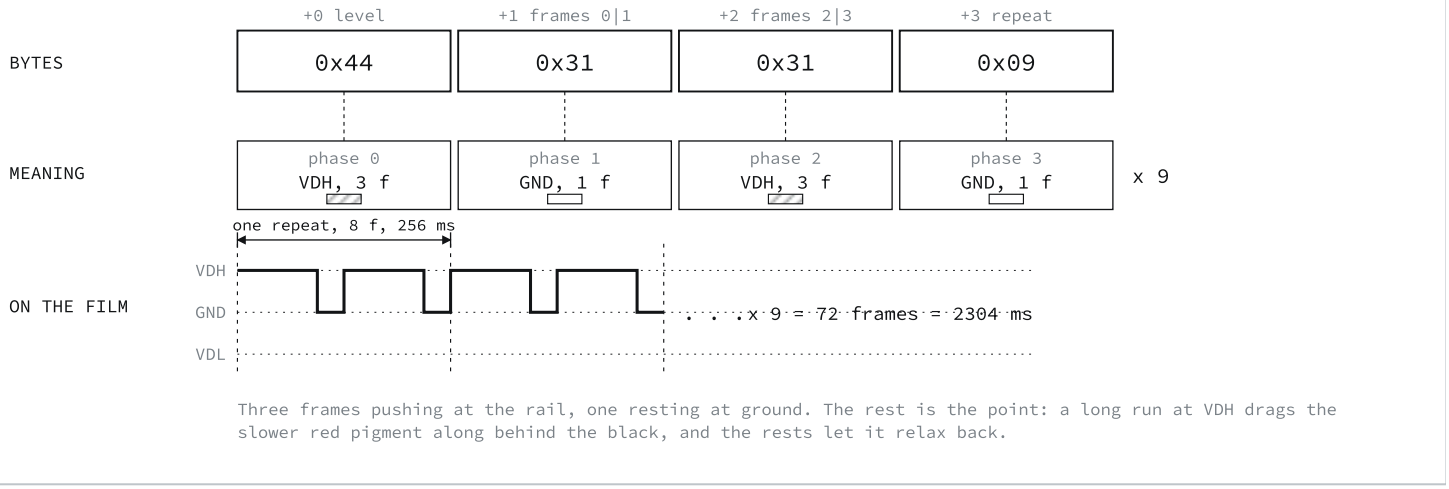
TABLE 2-4 LEVEL BYTE FIELDS AND RAIL CODES

BITS	FIELD	CODE	RAIL	EFFECT
7:6	phase 0	0	GND	no field. Not neutral over long stages: black relaxes to grey, white drifts
5:4	phase 1	1	VDH	full positive. Brings black up, and red with it
3:2	phase 2	2	VDL	negative. Drives towards paper
1:0	phase 3	3	VDHR	reduced positive. Above red's threshold, below black's

Hatch key: GND VDH VDL VDHR

2.5.2 Group record

FIG. 2-4 ONE GROUP RECORD DECODED



Four bytes become four phases and a repeat count. Frame counts are nibble-packed, high nibble first, so a single phase is capped at 15 frames.

2.5.3 Base table, complete listing

The table as written when the frame carries no red and the previous frame carried none. Tables 1, 2 and 3 are identical here; group 2 through group 8 are identical to each other.

TABLE 2-5 084 BASE TABLE, ALL 180 BYTES

TABLE	GROUP	BYTES	PHASE LEVELS	FRAMES	REP	DURATION	ROLE
0 paper	0	66 33 33 02	VDH VDL VDH VDL	3 3 3 3	2	24 f / 768 ms	balanced shake
0 paper	1	88 31 31 09	VDL GND VDL GND	3 1 3 1	9	72 f / 2304 ms	chopped drive to paper
0 paper	2–8	00 10 00 01	GND × 4	1 0 0 0	1	1 f / 32 ms	placeholder
1, 2, 3	0	99 33 33 02	VDL VDH VDL VDH	3 3 3 3	2	24 f / 768 ms	shake, opposite phase to paper
1, 2, 3	1	44 31 31 09	VDH GND VDH GND	3 1 3 1	9	72 f / 2304 ms	chopped drive to ink
1, 2, 3	2–8	00 10 00 01	GND × 4	1 0 0 0	1	1 f / 32 ms	placeholder
4 VCOM	0	00 33 33 02	GND × 4	3 3 3 3	2	24 f / 768 ms	held at ground, but carries the same frame counts
4 VCOM	1	00 31 31 09	GND × 4	3 1 3 1	9	72 f / 2304 ms	
4 VCOM	2–8	00 10 00 01	GND × 4	1 0 0 0	1	1 f / 32 ms	

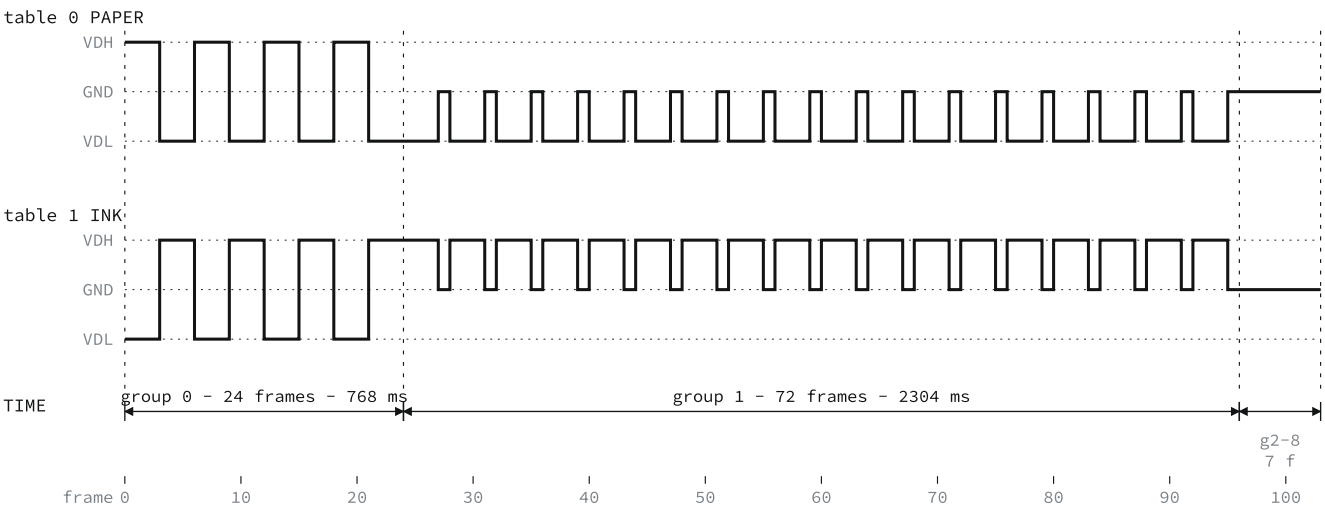
Total 103 frames, 3296 ms, against about 51 s for the panel's own waveform.

NOTE The seven placeholder groups are not padding that could be dropped. A group left at zero is not an empty group — it runs about four seconds.

Every table in a group carries the group's frame counts, VCOM included, because a group ends when the shortest table in it ends.

2.5.4 Frame-accurate trace

FIG. 2-5 ALL 103 FRAMES, PAPER AND INK



One step = one frame = 32.0 ms. 103 frames = 3296 ms. Table 4 (VCOM) stays at GND for all 103 frames but carries the same frame counts, because a group ends when the shortest table in it ends.

The two tables are mirror images through the shake, then separate: paper to VDL with a rest at ground, ink to VDH with the same rest. The shake is balanced — as much VDH as VDL — so it adds no net charge whatever a pixel holds. Without that the paper goes pink within a handful of updates.

2.5.5 Red variants

Red on this film is not a third direction. Both coloured pigments carry the same charge and both rise on a positive field; they differ only in threshold, red's being lower. On VDH black comes up with the red and has to be pulled back out, and the pull that removes black removes red with it — the two differ in speed by about 2.5, not the ten to one the literature suggests. So red is driven on VDHR, where black never comes up at all.

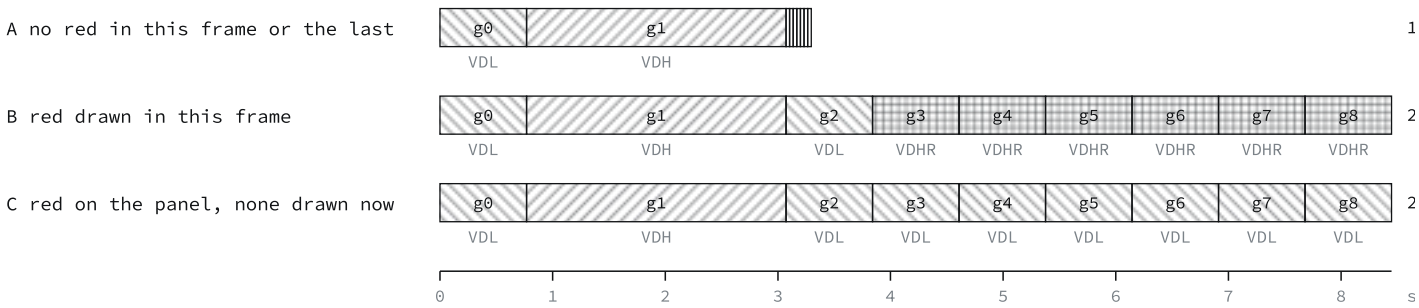
Groups 0 and 1 are never rewritten: putting the red stages over group 1 would replace nine repeats with one, which means grey blacks and red landing on an undriven pixel. The rewrite starts at group 2.

TABLE 2-6 WHAT THE REWRITE PUTS IN GROUPS 2 TO 8

VARIANT	TABLE	GROUP	BYTES	LEVELS	PURPOSE
B red drawn	0 paper	2–8	88 66 66 01	VDL GND VDL GND	held at its own target, not parked at ground
	1, 3 ink	2–8	44 66 66 01	VDH GND VDH GND	held at its own target
	2 red	2	88 66 66 01	VDL GND VDL GND	erase to white first
	2 red	3–8	FF 66 66 01	VDHR × 4	hold the reduced positive rail
C clearing	2 red	2–8	88 66 66 01	VDL GND VDL GND	all seven groups erase — there is red to remove and none to make
B and C	4 VCOM	2–8	00 66 66 01	GND × 4	ground, with the group's timing

Frame count `0x66` is six frames per phase, 192 ms. A rewritten group is therefore 24 frames, and the table grows from 103 to 264 frames — 8.45 s.

FIG. 2-6 THE RED TABLE PLAYED OUT IN TIME, THREE VARIANTS



In A the last seven groups are one frame each - placeholders, because a group left at zero is not an empty group, it runs about four seconds. In B and C the same seven groups carry 24 frames each, which is where the 5.15 s goes.

FIG. 2-7 ALL TABLES BY GROUP, THREE VARIANTS

A base - 103 frames, 3.30 s

	g0	g1	g2	g3	g4	g5	g6	g7	g8
0 paper	66	88	00	00	00	00	00	00	00
1 ink / 3	99	44	00	00	00	00	00	00	00
2 red	99	44	00	00	00	00	00	00	00
4 VCOM	00	00	00	00	00	00	00	00	00
frames x rep	12x2	8x9	1x1	1x1	1x1	1x1	1x1	1x1	1x1

B red drawn - 264 frames, 8.45 s

	g0	g1	g2	g3	g4	g5	g6	g7	g8
0 paper	66	88	88	88	88	88	88	88	88
1 ink / 3	99	44	44	44	44	44	44	44	44
2 red	99	44	88	FF	FF	FF	FF	FF	FF
4 VCOM	00	00	00	00	00	00	00	00	00
frames x rep	12x2	8x9	24x1	24x1	24x1	24x1	24x1	24x1	24x1

C clearing - 264 frames, 8.45 s

	g0	g1	g2	g3	g4	g5	g6	g7	g8
0 paper	66	88	88	88	88	88	88	88	88
1 ink / 3	99	44	44	44	44	44	44	44	44
2 red	99	44	88	88	88	88	88	88	88
4 VCOM	00	00	00	00	00	00	00	00	00
frames x rep	12x2	8x9	24x1	24x1	24x1	24x1	24x1	24x1	24x1

Cell = one group. Width inside a cell is split between the four phases in proportion to their frame counts; the hatch is the rail. Hex is the level byte. Groups 0 and 1 are never rewritten.

NOTE Published three-pigment waveforms carry an activation and a separation swing, and both rock on VDH — the rail that brings black up. On a red pixel that is exactly what must not happen, so neither appears here: the red table sees the negative rail and then the reduced positive one, and nothing else.

2.6 Shutdown

TABLE 2-7 084 SHUTDOWN

#	INDEX	BYTES	MEANING	DELAY AFTER
1	0x09	7F	booster off	20 ms
2	0x05	7D	panel setting	—
3	0x09	00	booster register cleared	200 ms
4	—	—	DC low, both chip selects low	150 ms
5	—	—	RST low, then VCC_EN released	300 ms
6	—	—	chip selects high, RST high. Rails bleed down from here	—

3 Driver B — E2969JS0B1

Complete sequence for this variant. Identified from the OTP field positions, not from a header byte. Waveform bank `0x84`, 400-byte short-waveform table, refresh duration fixed at about 1070 ms regardless of the table.

3.1 OTP probe

TABLE 3-1 B1 OTP READ

#	INDEX	BYTES	MEANING
1	—	—	VCC_EN on, 800 ms, then RST high 50 / low 100 / high 400 ms
2	`0x70`	`71, 36`	booster configuration. Sent to **both** COGs. A panel that never receives it draws about 2 A during refresh
3	`0xB9`	—	read opener, index only, master COG only — both COGs would otherwise drive the shared data line
4	—	`1 + 88 B`	SDA turned to input; one dummy byte, then the 88-byte block
5	—	—	identified if OTP[0x16] = `0x3B`, OTP[0x0D] = `0x3B` and OTP[0x28] bit 7 set
6	—	—	block remapped into the canonical layout — section 4.3

3.2 Initialisation sequence

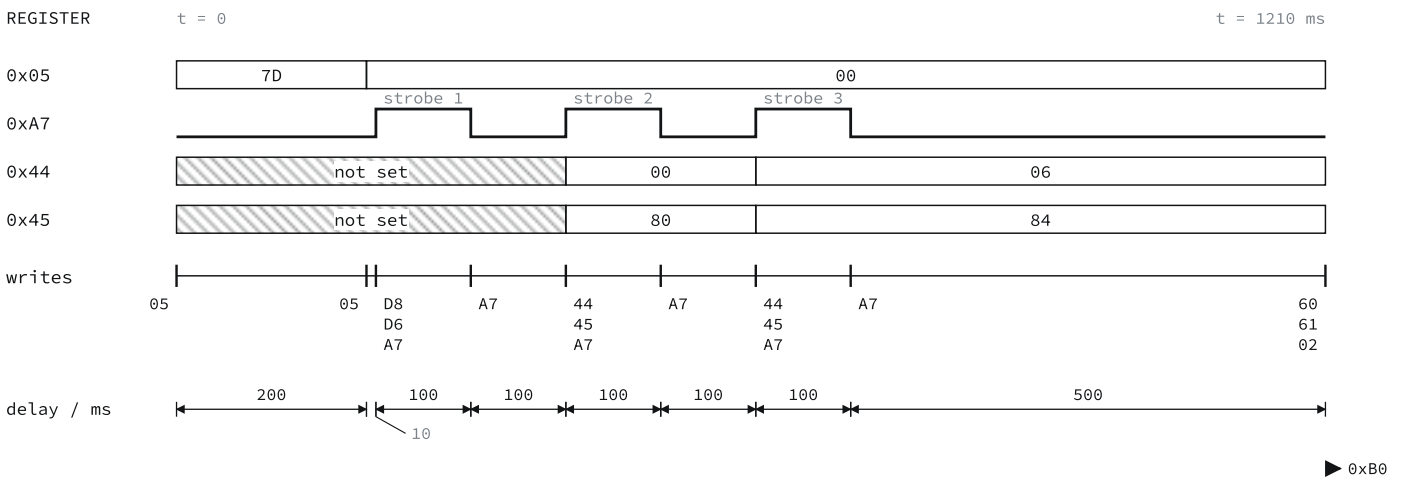
Execution order for one image. Two differences in position are worth marking before the table: DCTL is sent **before** the window registers rather than at the end of the register sequence, and there is no oscillator write at all.

TABLE 3-2 B1 INITIALISATION, COMPLETE

#	INDEX	BYTES	MEANING OF EACH BYTE	DELAY AFTER
POWER AND RESET				
1	—	—	VCC_EN asserted. The 1500 ms minimum supply-off time is waited out first	800 ms
2	—	—	RST high	50 ms
3	—	—	RST low	100 ms
4	—	—	RST high	400 ms
FRAME DATA				
5	0x01	OTP[10]	DCTL. **Sent here on this variant**, ahead of the window registers	—
6	0x13	OTP[16..1B]	DUW — display update window. 6 bytes, see Table 3-3	—
7	0x90	OTP[0C..0F]	DRFW — data refresh window. 4 bytes, see Table 3-3	—
8	0x12	OTP[13..15]	RAM_RW pointer. 3 bytes, forwarded verbatim	—
9	0x10	80 640 B	black/white plane. 672 lines × (60 B with CS1 low, 60 B with CS2 low)	—
10	0x12	OTP[13..15]	RAM_RW pointer again, before the second plane	—
11	0x11	80 640 B	red plane. Always sent; a frame with no red sends zeros	660 ms
REGISTER SEQUENCE — 1210 MS IN TOTAL				
12	0x05	7D	opens the sequence	200 ms
13	0x05	00	—	10 ms
—	0xC2	—	not sent on this variant	—
14	0xD8	OTP[1D]	unnamed panel setting, from the fuse block	—
15	0xD6	OTP[1E]	unnamed panel setting, from the fuse block	—
16	0xA7	10	load strobe 1, high	100 ms
17	0xA7	00	load strobe 1, low. **No oscillator write follows here**	100 ms
18	0x44	00	waveform source pointer	—
19	0x45	80	bank, the driver's own default	—
20	0xA7	10	load strobe 2, high	100 ms
21	0xA7	00	load strobe 2, low	100 ms
22	0x44	06	waveform source pointer	—
23	0x45	84	bank. 0x84 on this variant	—
24	0xA7	10	load strobe 3, high	100 ms
25	0xA7	00	load strobe 3, low. This pair commits the OTP waveform	500 ms
26	0x60	OTP[0B]	TCON, before STV	—
27	0x61	OTP[1C]	STV	—
—	0x01	—	not sent here — it went out at step 5	—
28	0x02	OTP[11]	closes the sequence	—
WAVEFORM, BOOSTER, TRIGGER				
29	0xB0	400 B	short waveform table — fast refresh only. Section 3.5	—
30	0x09 0x51	from OTP	DC/DC soft start. Four stage records at OTP[0x20..0x3F], section 4.2	100 ms
31	—	—	wait for BUSY high. This variant is usually **still ramping** here; triggering into a ramping booster is the "sits still, then suddenly draws" fault	—
32	0x15	3C	start the update. Runs ~1070 ms	—
33	0x15	3C	second pull. No plane data, no register touched	—

#	INDEX	BYTES	MEANING OF EACH BYTE	DELAY AFTER
34	0x15	3C	third pull	—
35	—	—	watch 1 s in case BUSY drops again for a second waveform pass	—

FIG. 3-1 B1 REGISTER SEQUENCE AGAINST REAL TIME



Same axis, same 1210 ms. `0xC2` and `0x01` are absent from the burst row and no `0x03` follows strobe 1; because neither omitted write carries a delay, the sequence takes exactly as long as it does on the other variant. The bank written before strobe 3 is `0x84`.

3.3 Window registers, byte by byte

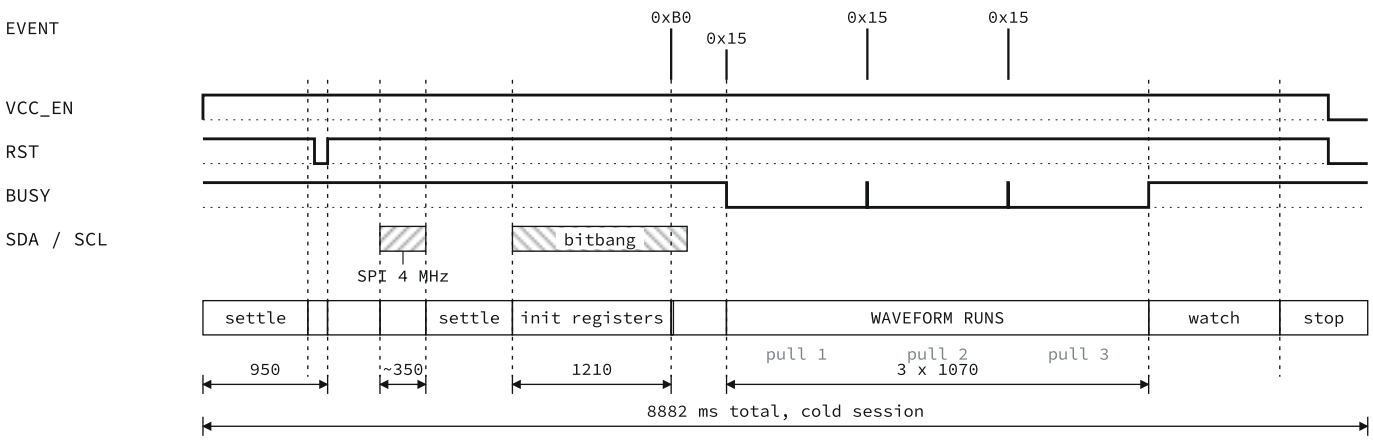
Identical fields and identical OTP offsets to the 084 — but only because the block was remapped during the probe. In the raw block these fields sit one byte lower from 0x12 onward; see section 4.3.

TABLE 3-3 MULTI-BYTE PAYLOADS, AFTER THE REMAP

INDEX	BYTE	CANONICAL	RAW B1	MEANING
0x13 DUW	[0]	OTP[0x16]	0x15	x start, in bytes
	[1]	OTP[0x17]	0x16	x end = 0x3B = 59, i.e. 60 bytes per COG. Also a detection byte
	[2]	OTP[0x18]	0x17	y window. Forwarded verbatim
	[3]	OTP[0x19]	0x18	
	[4]	OTP[0x1A]	0x19	
	[5]	OTP[0x1B]	0x1A	
0x90 DRFW	[0]	OTP[0x0C]	0x0C	forwarded verbatim
	[1]	OTP[0x0D]	0x0D	x end = 0x3B. Also a detection byte
	[2]	OTP[0x0E]	0x0E	forwarded verbatim
	[3]	OTP[0x0F]	0x0F	
0x12 RAM_RW	[0]	OTP[0x13]	0x12	RAM read/write pointer. Forwarded verbatim, re-sent before each plane
	[1]	OTP[0x14]	0x13	
	[2]	OTP[0x15]	0x14	

3.4 Refresh timing

FIG. 3-2 B1 FAST REFRESH, SUPPLY-ON TO SHUTDOWN



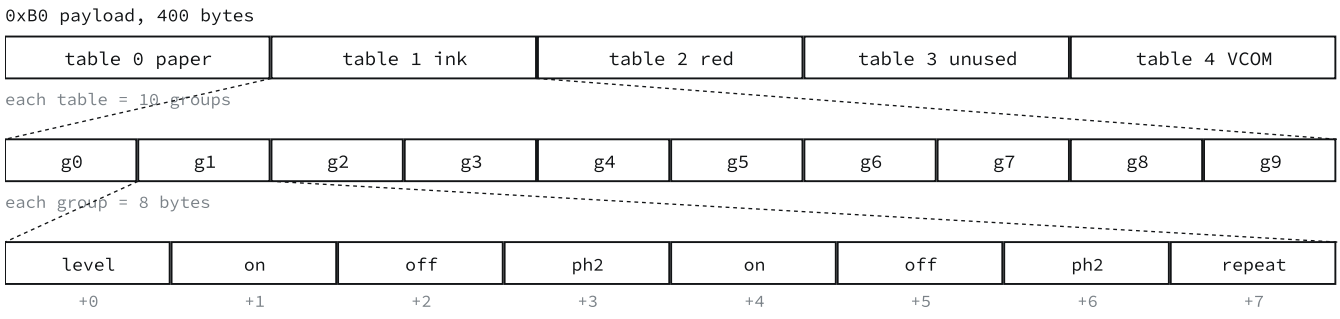
All figures in ms. A kept-open session re-enters at the plane write and ends at the trigger: no supply cycle, no reset, no register sequence, no soft start.

Three separate BUSY low periods of about 1070 ms each. Everything up to the trigger is loaded once; a pull is a bare `0x15`.

3.5 Waveform table at 0xB0

400 bytes: five tables of ten groups of eight bytes. One whole byte per frame count, so a phase is not capped at 15 the way the 084's nibbles cap it.

FIG. 3-3 400-BYTE PAYLOAD GEOMETRY

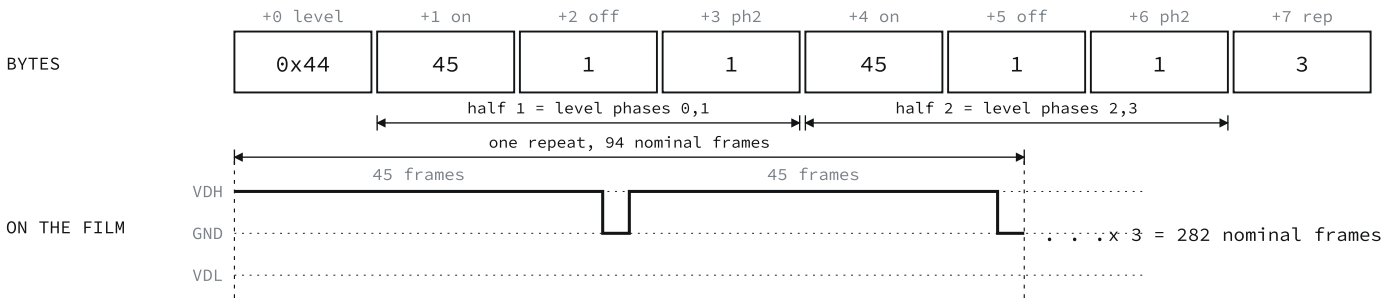


Byte offset of table t, group g, byte n = (t * 10 + g) * 8 + n

3.5.1 Group record

Level byte first, then two (on, off, ph2) triples, then the repeat count. The level byte uses the same encoding as Table 2-4: two bits per phase, phase 0 in the top bits.

FIG. 3-4 ONE GROUP RECORD DECODED



Every level this driver writes has the form AB AB, so phases 0 and 2 share a level and phases 1 and 3 share one. That matches the two identical triples: each is a drive followed by a rest.

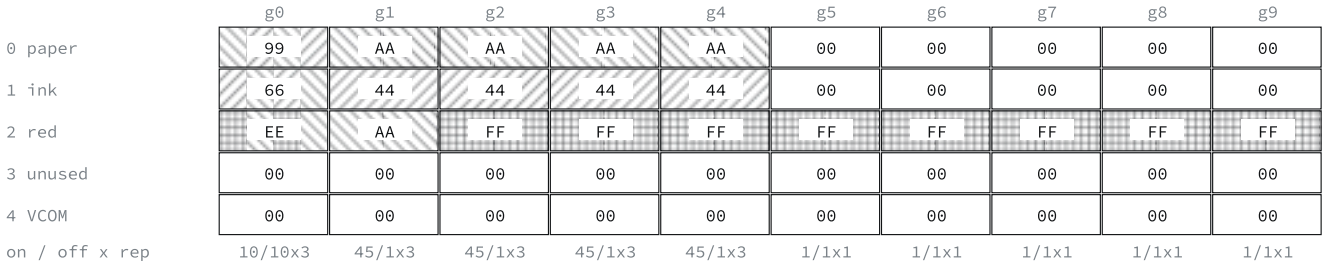
3.5.2 Table listing

TABLE 3-4 B1 TABLE, ALL 400 BYTES

TABLE	GROUP	BYTES	PHASE LEVELS	ON/OFF	REP	ROLE
0 paper	0	99 0A 0A 01 0A 0A 01 03	VDL VDH VDL VDH	10 / 10	3	split shake, one half of the pixels
0 paper	1–4	AA 2D 03 01 2D 03 01 03	VDL × 4	45 / 3	3	drive to paper, longer rest
0 paper	5–9	00 01 01 01 01 01 01 01	GND × 4	1 / 1	1	finished
1 ink	0	66 0A 0A 01 0A 0A 01 03	VDH VDL VDH VDL	10 / 10	3	split shake, opposite phase to paper
1 ink	1–4	44 2D 01 01 2D 01 01 03	VDH GND VDH GND	45 / 1	3	drive to ink
1 ink	5–9	00 01 01 01 01 01 01 01	GND × 4	1 / 1	1	finished
2 red	0	EE 0A 0A 01 0A 0A 01 03	VDHR VDL VDHR VDL	10 / 10	3	split shake
2 red	1	AA 2D 01 01 2D 01 01 03	VDL × 4	45 / 1	3	clear before setting — a pixel on its way to red may be coming from black
2 red	2–9	FF 2D 01 01 2D 01 01 03	VDHR × 4	45 / 1	3	reduced positive rail, eight groups of it
3 unused	0	00 0A 0A 01 0A 0A 01 03	GND × 4	10 / 10	3	unreachable table, but it carries the group timing
3 unused	1–4	00 2D 01 01 2D 01 01 03	GND × 4	45 / 1	3	
3 unused	5–9	00 01 01 01 01 01 01 01	GND × 4	1 / 1	1	—
4 VCOM	0–9	00 01 01 01 01 01 01 01	GND × 4	1 / 1	1	ground throughout

Hex for the counts: `0x0A` = 10, `0x2D` = 45, `0x03` = 3, `0x01` = 1.

FIG. 3-5 ALL FIVE TABLES BY GROUP



Tables 0 and 1 take one half of the pixels each, so they get opposite drives rather than the 084's one-white-table-three-black. The shake in g0 is split for the same reason: one level written into every table would swamp that division and push the whole panel one way.

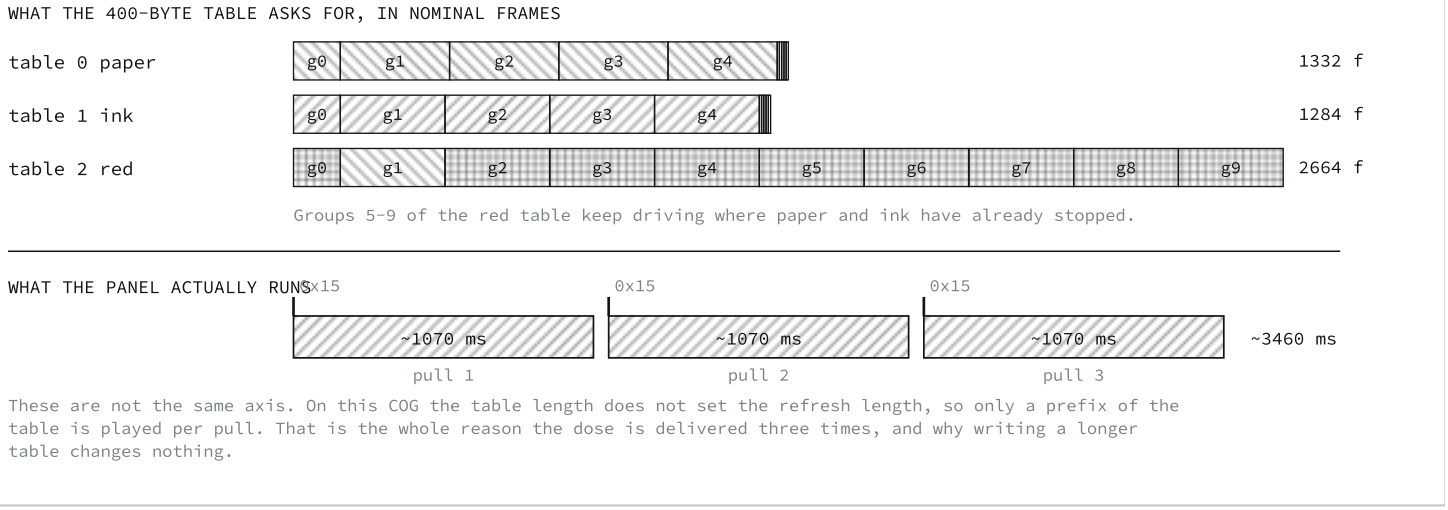
NOTE

Full duty on both tables costs paper brightness: 45/1 everywhere dropped it from L* 76.7 to 67.1, which is why paper keeps the longer 45/3 rest.

Measured against a conditioned film: ink at L* 15.1 with a* +1.88, paper at L* 86.9, contrast 71.8, in 3460 ms against 33 400 ms for the panel's own waveform.

3.5.3 Why the table is pulled three times

FIG. 3-6 NOMINAL TABLE LENGTH AGAINST THE REAL REFRESH WINDOW



The refresh runs for a fixed time whatever the table asks for, so a single pass cannot carry enough dose to clear the previous frame however long the table is written. The dose is delivered as three pulls of the already-latched waveform instead.

3.6 Shutdown

Identical to the 084: `0x09←7F`, 20 ms; `0x05←7D`; `0x09←00`, 200 ms; DC and both chip selects low, 150 ms; RST low and VCC_EN released, 300 ms; chip selects and RST high.

4 OTP block

88 bytes of fuse, read once and cached. It cannot change between refreshes, and the read costs about 2.8 s because it is clocked at ~50 kHz and both openers are tried.

4.1 Byte map

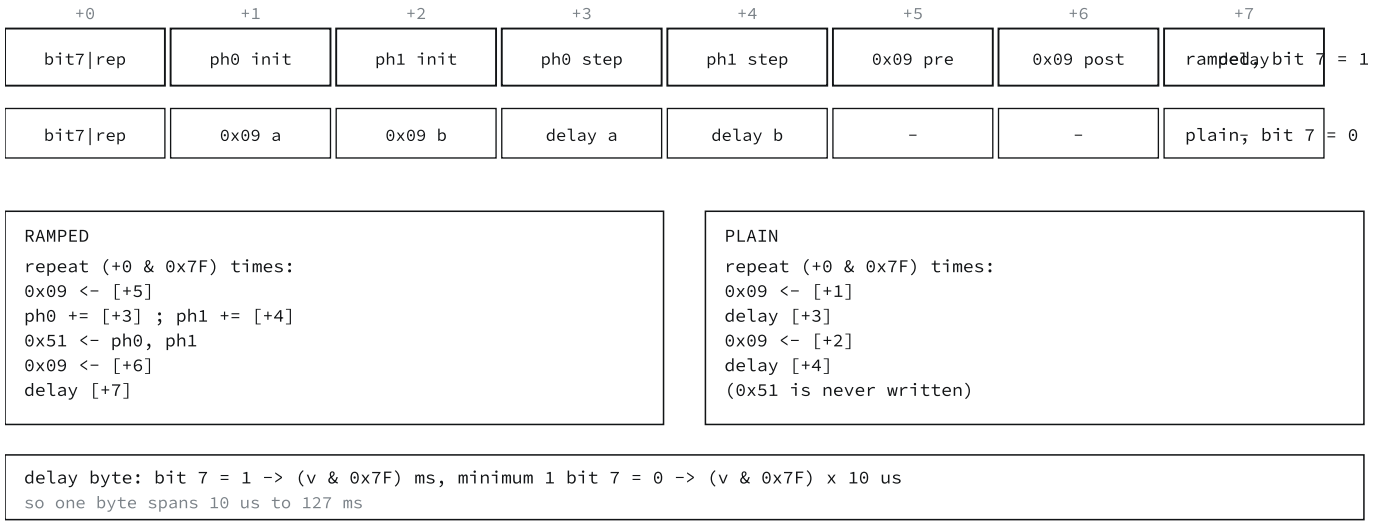
TABLE 4-1 FIELD MAP, BOTH LAYOUTS

CANONICAL	RAW B1	LEN	FIELD	GOES TO
0x00	0x00	1	header	must read 0xA5 on an 084; forced to 0xA5 after a B1 remap
0x01–0x0A	same	10	not used by this driver	—
0x0B	same	1	TCON	0x60
0x0C–0x0F	same	4	DRFW	0x90. Byte 0x0D validated as 0x3B
0x10	same	1	DCTL	0x01
0x11	same	1	unnamed	0x02
0x12	aliased	1	oscillator	0x03, second byte. 084 only
0x13–0x15	0x12–0x14	3	RAM_RW	0x12, before each plane
0x16–0x1B	0x15–0x1A	6	DUW	0x13. x end validated as 0x3B
0x1C	0x1B	1	STV	0x61
0x1D	0x1C	1	unnamed	0xD8
0x1E	0x1D	1	unnamed	0xD6
0x20–0x3F	0x28–0x47	32	DC/DC ramp	four 8-byte stage records, section 4.2

4.2 DC/DC ramp stage record

Four records of eight bytes. Bit 7 of the first byte selects the loop shape; the low seven bits are the repeat count.

FIG. 4-1 RAMP STAGE RECORD AND THE TWO LOOP SHAPES

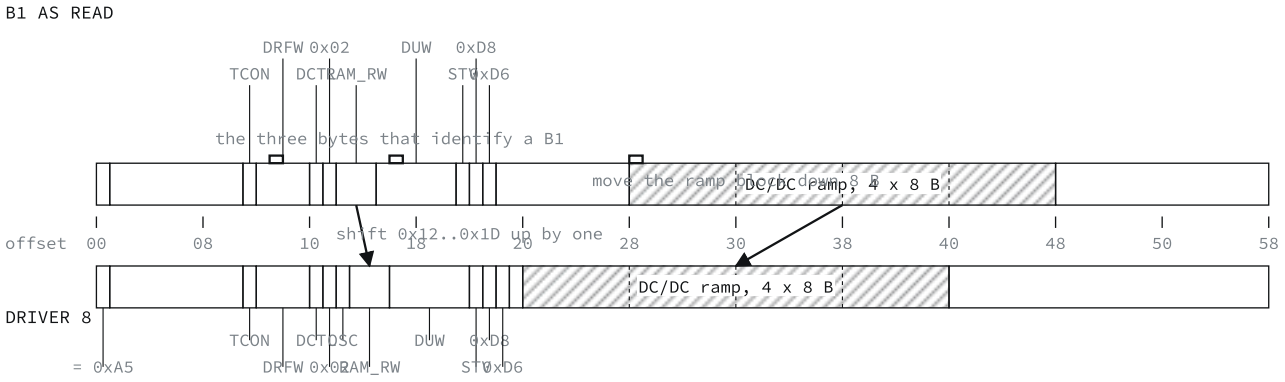


The repeat count is run verbatim, with no clamp. 0xFF asks for 127 rounds and hangs BUSY, which is why all four stages are rejected up front rather than fixed up here.

NOTE All four stages are validated before the booster is driven: a stage byte must be neither `0xFF` nor zero repeats nor more than 40. `0xFF` asks for 127 ramp rounds and hangs BUSY. Changing how the booster ramps is what stresses a panel, so a garbled block is refused rather than clamped.

4.3 Driver-B remap

FIG. 4-2 THE TWO MOVES THAT CONVERT A RAW B1 BLOCK



Then OTP[0] is forced to 0xA5, so everything downstream reads one layout. Native 0x12 ends up in both OSC and RAM_RW[0]; harmless, because 0x03 is never sent on a B1.

5 Refresh control

5.1 Mode and red handling

A fast refresh is black and white. The short waveform cannot make a good red: on the 084 the red table is barely reached in the time the short table runs, and on the B1 the fixed window leaves red pale and getting worse with every frame that half-drives it. Red content is drawn as ink instead of being dropped, so the picture stays readable and simply goes monochrome until the next slow refresh.

Red already on the film is the harder case. A few clearing groups are not enough — measured on the panel the red survives them, and once it has sat through one fast frame it does not come out of the next one either. So the first fast refresh after a red frame silently runs on the panel's own waveform. The promotion is to the waveform only; red stays suppressed, which is also what stops it repeating.

FIG. 5-1 REFRESH DECISION FLOW



TABLE 5-1 WHAT EACH CASE ACTUALLY DRIVES

CASE	WAVEFORM	RED DRAWN	084 TABLE	B1 PULLS
slow, no red	OTP	—	not used	1
slow, red	OTP	yes	not used	1
fast, no red anywhere	0xB0	—	base, 103 f	3
fast, red in this frame, default config	0xB0	no, as ink	base, 103 f	3
fast, red in this frame, red allowed	0xB0	yes	variant B, 264 f	3
fast, red still on the panel	OTP, promoted	no	not used	1

5.2 Session and latching

The panel latches one waveform per session and will not take a second one. Rewriting `0xB0` inside an open session does not re-latch it — it drops the panel back onto its own waveform. Anything that needs a different table therefore closes the session first: a change of mode, or on an 084 a change in whether the table carries red.

With the session held open, a later image costs only its plane data and a bare `0x15`: no supply cycle, no reset, no register sequence, no soft start. The session is torn down every 20 images regardless.

6 Timing constants

TABLE 6-1 MEASURED AND MANDATED TIMINGS

STEP	VALUE	WHY
Minimum supply-off time	1500 ms	The booster rails have to bleed down. At 800 ms the COG init fails and the OTP reads back corrupt
Settle after supply on	800 ms	Vendor reset shape with margin added
Reset pulse	50 / 100 / 400 ms	High, low, high. Shortening it shows up as a failed init, not a worse picture
OTP read	~2800 ms	88 bytes at ~50 kHz, both openers tried. Paid once per power cycle
Plane transfer	~350 ms	2 × 80 640 B at 4 MHz. Bitbanging this is what made a frame eighteen seconds
Settle before the register sequence	660 ms	—
Register sequence	1210 ms	200 + 10 + 6 × 100 + 500, identical on both variants
Soft start	from the OTP	Four stages, repeat counts run verbatim
Post-soft-start settle	100 ms + wait	Ceiling 10 s on BUSY, then it proceeds anyway
Refresh, own waveform	~51 s / ~34 s	084 / B1
Refresh, short waveform	3296 ms / ~3460 ms	084 one trigger / B1 three pulls
Second-pass watch	1000 ms	Some panels drop BUSY again for a second waveform pass
Idle ceiling	180 000 ms	Has to clear a 51 s refresh with room to spare
Shutdown	670 ms	20 + 200 + 150 + 300

7 Errata

1. **Supply polarity does not fail like a power fault.** With the enable inverted the OTP read still succeeds, because it runs during a window where the pin happens to sit at the other level. The failure appears only at the soft start, as a booster that never reports ready.
2. **`0xB0` has one window.** Before the last load strobe it is overwritten by the OTP waveform; after the soft start it is too late; inside an open session it does not re-latch.
3. **A zero group is not an empty group.** It runs about four seconds.
4. **Group timing applies to every table.** A table left at one frame of ground ends the group there whatever the others asked for, VCOM included.
5. **`EPD_T_PRE_SS` is in units of 10 ms**, unlike every other timing constant in the configuration header. The value 50 produces 500 ms, which is what both sequence tables show.
6. **Both plane bits set lands on table 3**, which no waveform drives.
7. **"Red is on the panel" has to outlive RAM.** The film is not reset by the MCU losing its memory, and a wake that believes the panel is red-free will run a fast refresh straight over red that nothing short can remove. The flag is kept in RTC memory.
8. **Do not release the bus pins while the panel is off.** Sound in theory — driving an unpowered COG backfeeds it through the ESD diodes — and it broke this panel: the refresh reached the soft start and BUSY never came ready. Only the two shared bus pins change hands between bitbang and SPI.
9. **On a B1, do not skip `0x70`.** Without it the panel draws about 2 A during refresh.

Derived from `src/epd97.cpp`, `src/epd97_lut.h` and `include/epd97_config.h`, itself a port of the EFR32FG22 OEPL driver `oepl_display_driver_pervasive_itc_medium.c`, which is the origin of the measured figures. Where the driver source does not evidence a register's function, this note describes its effect rather than assigning a name.